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Palaeomagnetism of the West Sakhalin Basin: evidence for northward displacement during the Cretaceous

Alexandra Abrajevitch,[1,*] Sergey Zyabrev,[2] Alexei N. Didenko[2] and Kazuto Kodama[1]

[1]*Center for Advanced Marine Core Research, Kochi University,* B200 *Monobe Nankoku, Kochi,* 783–8502*, Japan. E-mail: alexandra.abrajevitch@gmail.com*
[2]*Institute of Tectonics and Geophysics, Far Eastern Branch, Russian Academy of Sciences,* 65*, Kim Yu. Chen Street,* 680063 *Khabarovsk, Russia*

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SUMMARY

A significant margin-parallel translation of terranes is postulated by all models for tectonic evolution of the East Asian continental margin, although the timing and magnitude of displacements of individual elements are poorly constrained as yet. The West Sakhalin Basin—a forearc basin associated with the Cretaceous volcanic arcs—is one of the displaced tectonic elements. Our palaeomagnetic study of the basin rocks provides the first quantitative constraints on its Cretaceous history. We identified a characteristic magnetization in the Berriasian–Valanginian basalt representing the oceanic basement of the basin and four characteristic magnetizations in a continuous sequence of the Albian to Maastrichtian siliciclastic fill of the basin. A combination of the positive fold, conglomerate and reversals tests supports the primary origin of the remanence in the sediments. Palaeomagnetic data indicate a gradual shift of the West Sakhalin Basin from subequatorial latitudes during the Early Cretaceous to about 40°N by the Late Cretaceous. The main phase of the margin-parallel migration took place during the Early Cretaceous, and ended by the Cenomanian. The continuous sedimentary sequence records ∼50 degrees of progressive clockwise rotation during the Late Cretaceous.

Key words: Palaeomagnetism applied to tectonics; Continental tectonics: strike-slip and transform; Asia.

1 INTRODUCTION

The East Asian continental margin has been formed by a complex combination of accretion and collision processes accompanied by strike-slip displacement of individual terranes and their assemblages (e.g. Zonenshain *et al.* 1990; Maruyama *et al.* 1997; Tazawa 2004). A significant area of the Russian Far East region is built up of accretionary complexes that formed along the convergent margin during the Cretaceous (Fig. 1). A complex distribution of coeval complexes as well as juxtaposition of different age units indicates a considerable margin-parallel translation of terranes, which was accommodated by sinistral strike-slip fault systems in the region (e.g. Otsuki 1992; Shengor & Natal'in 1996; Khanchuk 2001; Tazawa 2004).

Palaeogeographic reconstructions of the Asian margin mainly rely on structural information, such as strike-slip offset of the initially coherent bodies, to estimate the post-accretionary displacements; the estimated displacements are then 'reversed' and the initial configurations of the tectonic units are thus restored (e.g. Natal'in 1993; Zyabrev *et al.* 2005). For those terrane boundaries of which are exposed and accessible for inspection, the structural estimates of the post-accretionary displacements are easily available. When structural correlation is unfeasible, for example, because of the uncertain terrane affinity, only palaeomagnetism can provide the necessary constraints on the timing and magnitude of the displacements.

The West Sakhalin Basin (WSB)—a Cretaceous forearc basin—is one of the key tectonic elements of the Asian margin. The basin lies approximately opposite to its coeval Cretaceous accretionary complexes of mainland Asia, essentially sealing the inferred Cretaceous convergent margin (Fig. 1). Such juxtaposition suggests that the WSB was translated northwards to its present position some time after the accretion of the mainland complexes. The timing and the magnitude of the translation, however, is poorly constrained, and the direct structural estimates for this terrane are unavailable because its boundaries are hidden under the Tatar Strait. Here we report the results of a palaeomagnetic study of the WSB rocks that provide the first quantitative constraints on the location of the basin during the Cretaceous.

2 GEOLOGICAL SETTING

The WSB is considered as a forearc basin associated with the Late Cretaceous to Paleogene Eastern Sikhote-Alin volcanic belt as well as with an Early Cretaceous island arc fragments of which are found in Sikhote-Alin (Fig. 1). The basin continues to the south in Hokkaido Island, where the forearc deposits are known as the

*Now at: Research School of Earth Sciences, Australian National University, 61 Mills Street, Acton, ACT 0200, Australia.

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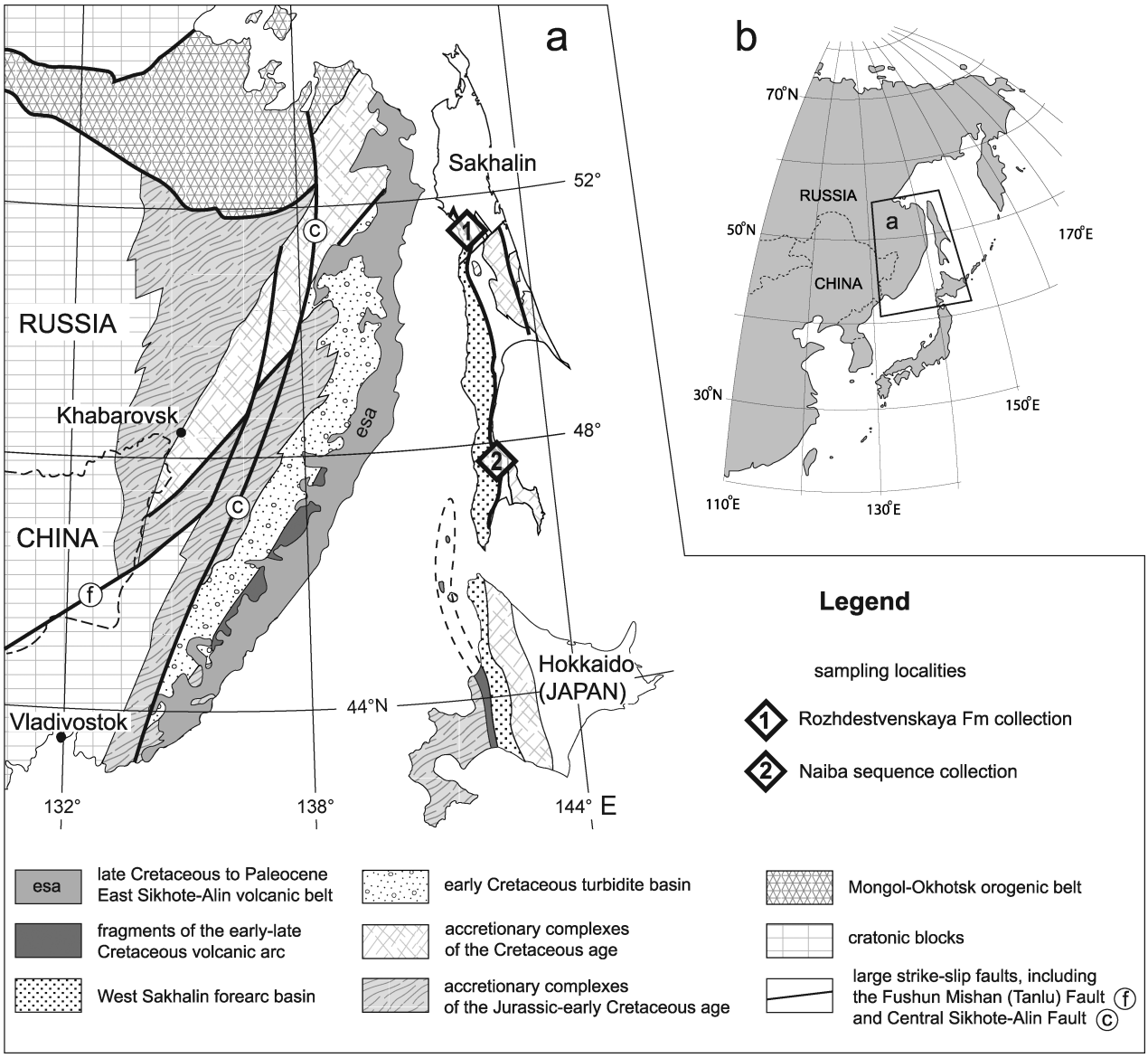


Figure 1. (a) Schematic tectonic map of the study area [modified after Natal'in (1991), Kiminami *et al.* (1992)] with sampling localities. Inset map (b) shows location of the study within north–east Asia.

Yezo Group. Unlike the Cretaceous strata in Hokkaido that generally occur as disrupted sequences at a single locality, those in South Sakhalin crop out continuously in the Naiba River area. There, the Cretaceous-age deposits are exposed within a west-dipping monocline with the dip angles on average 30–50° (Fig. 2). The monocline is locally complicated by folds, for example, at the confluence of Krasnoyarka and Naiba rivers, as well as by a series of small-scale normal and reverse faults (Fig. 2).

The Cretaceous rocks at the Naiba River area (Naiba sequence) comprise a ∼5000-m thick sequence of terrigenous rocks, alternating between predominantly fine-grained, clay–mudstone varieties, and coarse-grained sandstone layers (Vereshschagin 1977; Salnikov & Tikhomolov 1987); the sequence is interpreted as the result of deposition by turbidity currents (Zyabrev 1984; Zyabrev 1992). The rocks contain abundant fossils of inoceramid and other bivalves, ammonites, gastropods, foraminifera and radiolarian (Salnikov & Tikhomolov 1987 and references therein). The rich fossil assemblages allowed for the development of the detailed biostratigraphic zonation. A magnetostratigraphic study of the upper part of the sequence (Kodama *et al.* 2000) further refined dating of the rock units. Based on lithologic variations and the fossil content, the sequence was subdivided (from top to bottom) into the Krasnoyarka, Bykov, Naiba and Ai formations (Fig. 2). The formations were further subdivided into members and horizons.

The Krasnoyarka Fm (Late Campanian–Danian) is composed of sandstone and tuffaceous sandstone with thin layers of mudstones and rare conglomerate beds; its total thickness is 800–900 m. The upper part of the formation is characterized by a significantly higher content of tuffaceous material and contains a distinct fossil assemblage; accordingly, the formation was subdivided into the upper and lower members.

The Bykov Fm (Late Cenomanian–Early Campanian) is dominated by uniform clay- and mudstone with rare thin layers of turbiditic sandstone. The total thickness is about 2000 m. The formation is subdivided into three members based on minor lithological

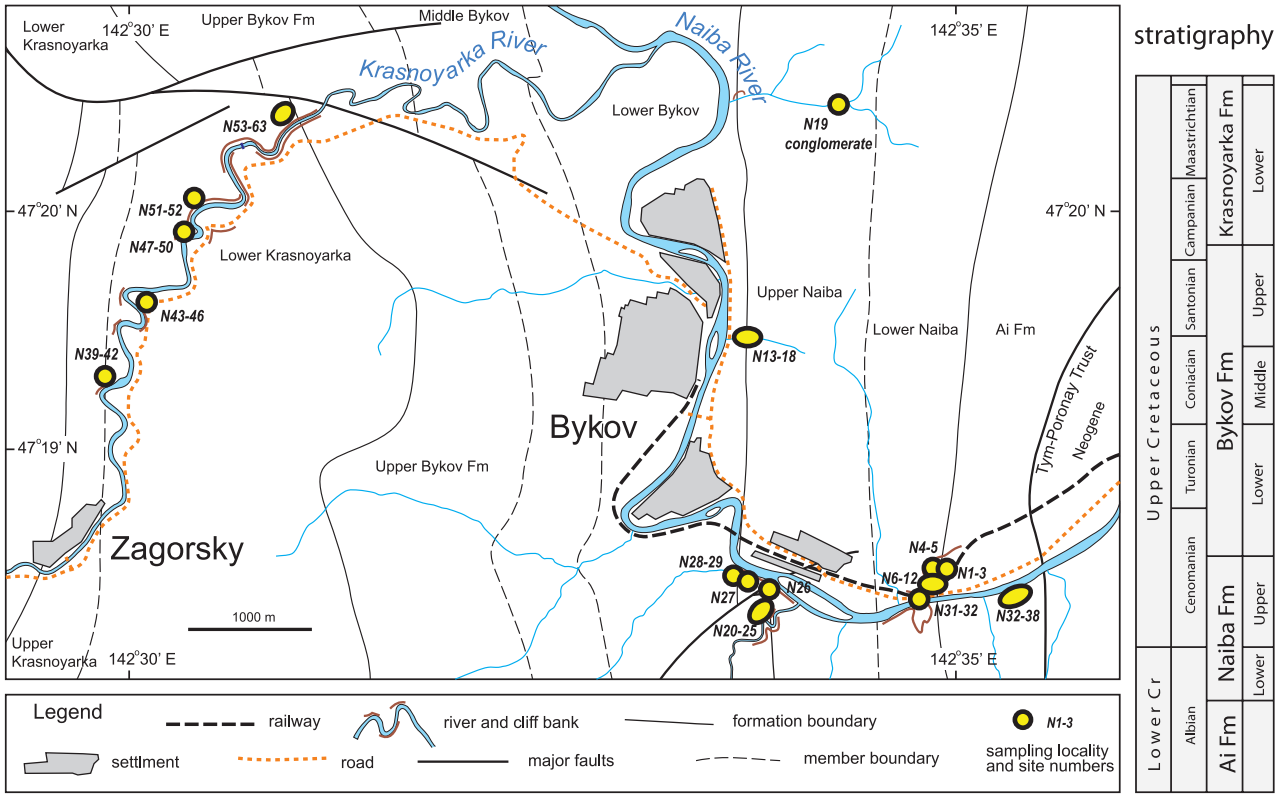


Figure 2. Map of Naiba area (GPS coordinates of Bykov N 47°19′ E142°33′) with sampling sites' locations (left); and stratigraphic sequence of the Naiba sequence (according to Vereschagin *et al.* 1987).

variations (the middle member is slightly more sandy than the upper and lower members).

The Naiba Fm (Late Albian–Early Cenomanian) is dominated by layers of massive sandstones separated by intervals of alternating sandstones and mudstones with rare layers and lenses of conglomerates. The thickness of the Naiba Formation is 1100–1300 m. The formation is subdivided into two members based on the relative abundance of sandstones (higher in the upper part) and distinct fauna; the fossil assemblage of the upper member of the Naiba Fm is similar to that of the lower part of the Bykov Fm.

The Ai Fm (Early Albian age) is characterized by alternation of thin (5–10 to 50–150 cm) layers of sandstones, claystones and mudstones. Its visible thickness is ∼580 m. The base of the formation is truncated by the Tym-Poronay thrust of Cenozoic age.

Rare outcrops of the basal part of the basin are known only in the northern Sakhalin, at the source of river Rozhdestvenka (Fig. 3), where they are exposed within imbricated sheets in the front of the Tym-Poronay thrust (Zyabrev 1992). There, the oldest deposits of the basin are represented by mudstones containing Early Aptian radiolarians (Zyabrev *et al.* 2004). The mudstones are in a depositional contact with underlying cherts and thin basalts of the Rozhdestvenka Fm that represents an oceanic basement of the basin. The age of the basalts is constrained by the radiolarian assemblages from the overlying chert as the Berriasian–Valanginian (Zyabrev & Bragin 1987).

3 SAMPLING

In the sedimentary Naiba sequence, we sampled sandstone beds during summer 2009 field season. We considered that each individual sandstone bed, being deposited during a single turbidite emplacement event, represents a spot reading of the field, that is, can be regarded as a site. Four–seven (on average six) oriented samples were collected from each site (sandstone bed); in total, we collected 355 samples at 61 sites. The sampling locations are shown in Fig. 2. The age of palaeomagnetic samples was assigned on the basis of their stratigraphic positions.

In addition, we collected 32 clasts from an intraformational conglomerate for a conglomerate test. The clasts vary in lithology; predominantly, we collected andesites and chert pebbles. Clasts of sandstones and mudstones similar in composition to the host sediments were also present, but those pebbles were very friable and we were unable to extract them.

We also studied basalt samples of the Rozhdestvenka Fm collected during the reconnaissance survey in 1984 by S. Zyabrev. A geological sketch of the locality, with the sampling sites indicated, is shown in Fig. 3. The samples were previously cut into standard specimens and stored. In total, 29 individually oriented basalt samples collected from nine flows were available for this study.

All samples were collected as oriented blocks; a magnetic compass with an inclinometer was used for sample orientation. A correction for the local magnetic deviation of 10°23′ W has been made to all relevant measurements in the Naiba area; a correction of 11°W has been made to the measurements at the Rozdestvenka locality. Block samples were subsequently cut into standard specimens—cubes with ∼20-mm side dimensions.

4 METHODS

Palaeomagnetic and rock magnetic measurements were performed in the Paleomagnetism Laboratory at the Center for Advanced Marine Core Research, Kochi University (Japan). Measurements

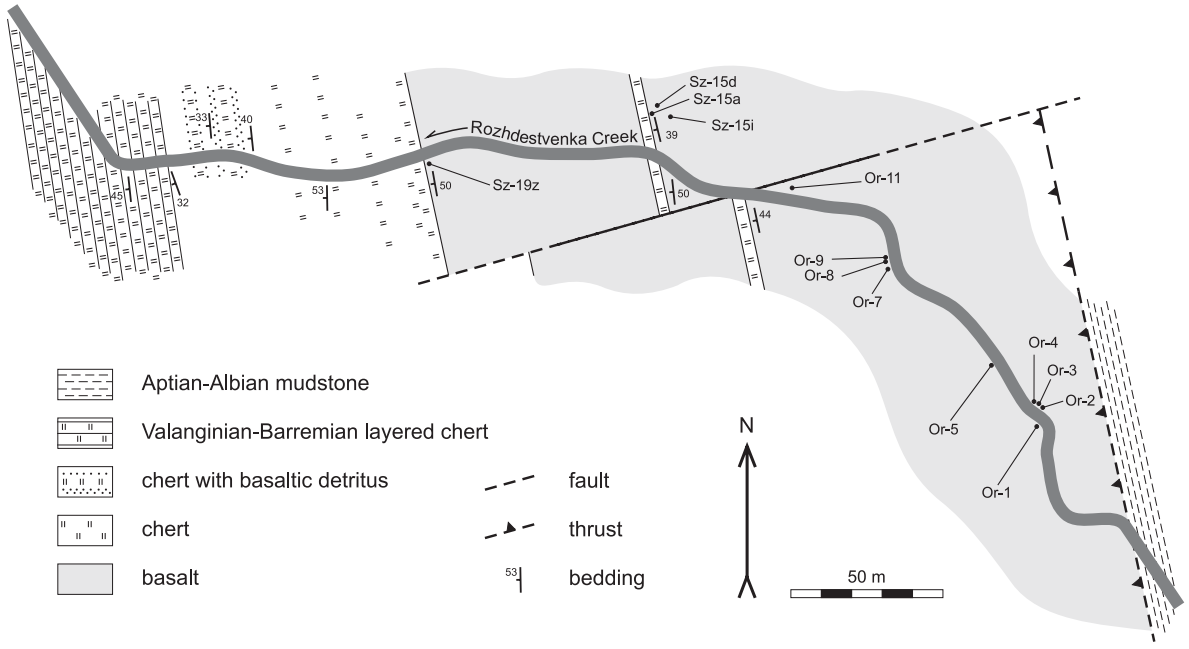


Figure 3. Geological sketch of the Rozhdestvenskaya sampling area (GPS coordinates N50°59′17″ E142°22′51″) with sampling sites indicated.

of natural remanent magnetization (NRM) were performed in a magnetically shielded room using a three-axis 2G superconducting magnetometer.

Representative pilot specimens of different rock types were subjected to alternating field (AF) and thermal demagnetization. AF demagnetization failed to isolate components of magnetization successfully in all samples. The pilot samples of basalts and conglomerate clasts tolerated stepwise heating well, and the rest of the samples from these groups were demagnetized thermally.

Only about a third of the pilot samples selected from the Naiba sequence collections survived thermal demagnetization, whereas the rest had acquired spurious magnetization at temperatures above 400 °C. Given that neither AF nor thermal demagnetization was successful in isolating magnetization components on its own, we tested a composite technique—thermal demagnetization up to 350–400 °C, followed by AF demagnetization. The results of the composite technique and thermal demagnetization only were identical (Fig. 4), demonstrating that these two methods are equally effective in isolating magnetization components. To avoid chemical alteration at high-temperature demagnetization steps, we used the composite demagnetization technique for the bulk of the Naiba sequence samples. Results of the demagnetization treatments have been plotted in orthogonal vector endpoint diagrams (Zijderveld 1967) and in stereonets. For calculation of the magnetization directions, a principal component analysis (Kirschvink 1980) was used on linear segments of the Zijderveld plots.

To identify the cause of the unstable demagnetization behaviour, we subjected several representative samples of the Naiba sequence rocks to a set of rock magnetic measurements. Thermomagnetic curves were acquired with a Curie balance (Natsuhara Giken); operational field 0.5 T. The samples were progressively heated in air to 700 °C at a rate of 10 °C min^{-1}, and then cooled to room temperature (RT) at the same rate.

Zero field cooled (ZFC) and field cooled (FC) remanence was measured for representative samples with a Quantum Design (San Diego, CA, USA) SQUID magnetometer [Magnetic Property Measurement System (MPMS), sensitivity of $\sim 10^{-7}$ Am^2]. ZFC

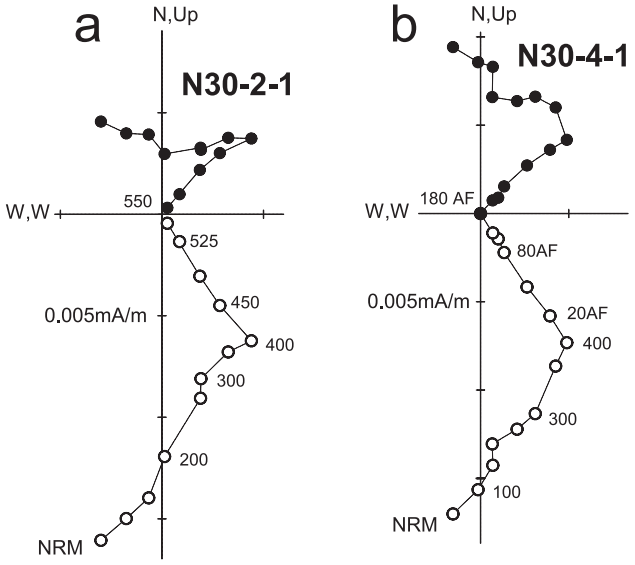


Figure 4. Orthogonal demagnetization diagrams (Zijderveld 1967) in tilt-corrected coordinates for a sample subjected to thermal demagnetization only (a) and a sample subjected to the composite treatment (b). Open symbols correspond to magnetization endpoints projected onto the vertical plane; closed symbols are projections onto the horizontal plane. Tick-marks denote intensities as labelled in mA m^{-1}. Numbers next to endpoints indicate demagnetization treatment; Alternative Field demagnetization steps (in mT) are indicated by AF symbols; numbers without AF symbol indicate thermal treatment in °C.

remanence was obtained by initial cooling from 300 to 20 K in zero field before the application of a 2.5-T field at 20 K. FC remanence was obtained by cooling from 300 to 20 K in a high magnetic field of 2.5 T. Changes in isothermal remanent magnetization (IRM) acquired in 2.5-T field at RT was monitored with the MPMS on cooling to 20 K and subsequently warming back to RT.

To test for the possible compaction-induced reorientation of the ferromagnetic carriers, and hence, the likelihood of palaeomagnetic

inclination shallowing, we measured anisotropy of anhysteretic remanence (AARM) (Jackson *et al.* 1988) for 16 samples representative of 8 sites of the Ai Formation. Each specimen was initially demagnetized in an AF of 120 mT. An anhysteretic remanent magnetization (ARM) was then imparted to the specimens at a specified orientation in the AF field of 120 mT with a superimposed DC field of 100 μT, using a Natsuhara Giken AF demagnetizer. The intensity of the ARM acquired was measured with a Natsuhara Giken (Osaka, Japan) spinner magnetometer. The procedure was repeated for nine different orientations for each specimen. These nine ARMs were used to determine a best-fitting anisotropy ellipsoid. The MS Excel program ARManiso provided by the Institute for Rock Magnetism, University of Minneapolis, was used for calculations.

5 RESULTS

5.1 Rozhdestvenka basalts (RB)

The basalt samples display a variety of demagnetization behaviours (Fig. 5). Two magnetization components are present in the majority of the samples. The low temperature component (LTC) has a range of the unblocking temperatures, from as low as 200 °C (e.g. Figs 5c and e) to more than 450 °C (e.g. Figs 5b, g and h). Directions of the LTC are widely scattered and likely represent viscous magnetization acquired during sample preparation and storage. After removal of the LTC, remanence usually decays univectorially to the origin. In a few samples, for example, Fig. 5(c), there seems to be a change

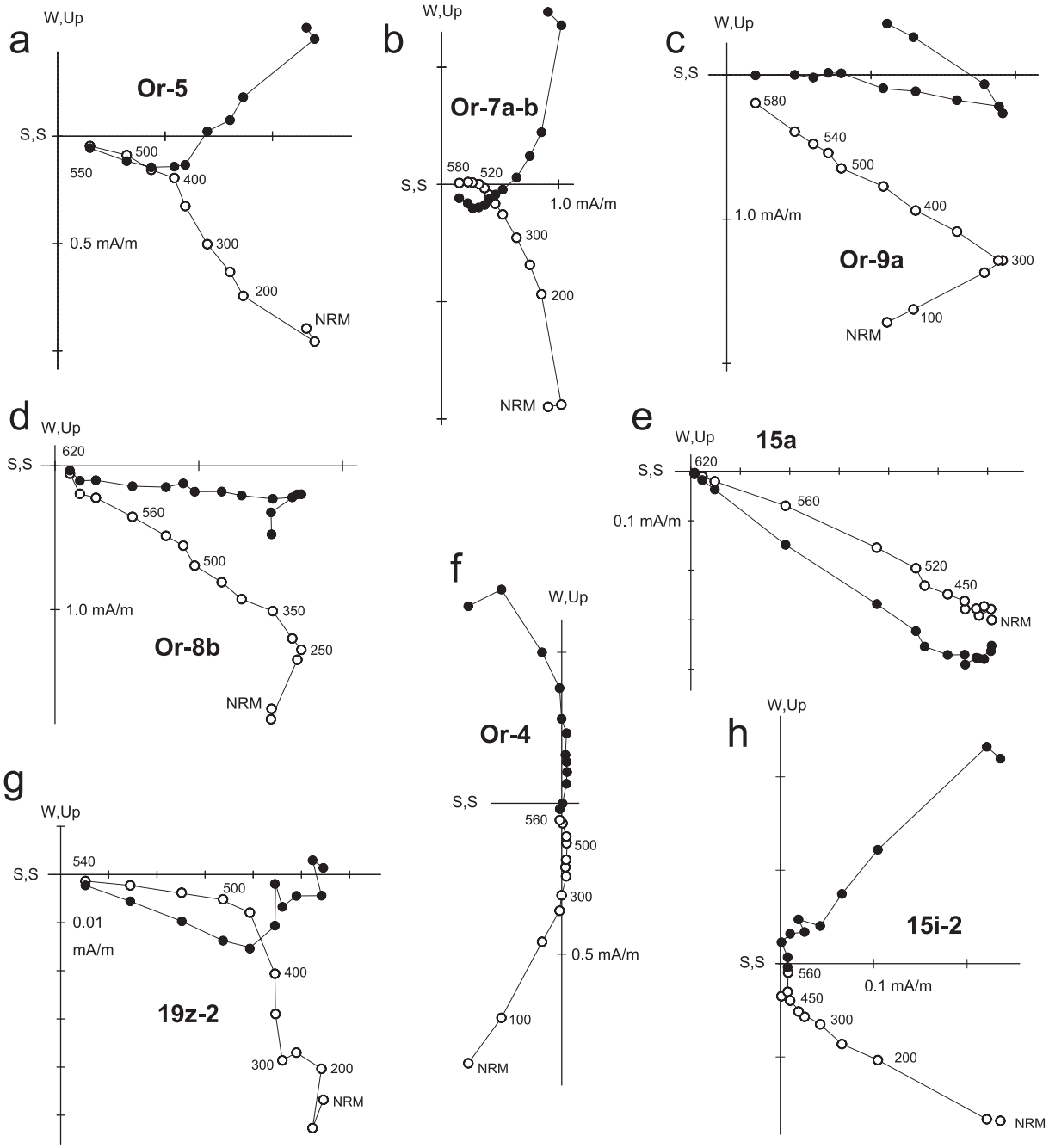


Figure 5. Orthogonal demagnetization diagrams in tilt-corrected coordinates for representative samples of the Rozhdestveskaya basalts (RB) collection. Plotting conventions as in Fig. 4.

Table 1. Characteristic remanence directions of the Rozdestvenka Fm (**RB** collection).

Site	Location		Bed	N	*In situ*			Tilt-corrected			VGP			
	Lat (N)	Long (E)			Dec	Inc	k	α_{95}	Dec	Inc	Lat (N)	Long (E)	dp	dm
Or-1/4	50.99	142.38	179/44	2/4	23.8	12.5	55.1		25.7	7.4	38.1	289.0		
Or-5	50.99	142.38	179/44	1/1	21.3	−12.3			23.6	6.0	38.1	291.9		
Or-7	50.99	142.38	179/44	3/3	34.4	−22.5	50.2	17.6	39.8	5.6	31.4	273.9	8.9	17.7
Or-8/9	50.99	142.38	179/44	4/4	29.9	17.7	96.6	10.9	8.8	33.9	56.8	307.0	7.1	12.4
Or-11	50.99	142.38	179/44	1/1	20.6	5.0			11.4	18.5	47.5	305.6		
Sz-15a	50.99	142.38	177/39	4/4	35.6	−6.5	194.5	6.6	32.5	17.6	40.2	278.3	3.5	6.8
Sz-15d	50.99	142.38	177/39	4/4	35.8	−13.7	473.8	4.2	36.4	11.4	35.6	275.8	2.2	4.3
Sz-15i[a]	50.99	142.38	177/39	5/5	35.4	72.3	15.2	20.3	294.8	59.2	44.6	64.9	22.7	30.4
Sz-19z	50.99	142.38	181/56	3/3	19.3	−13.1	154.1	10.0	22.3	7.3	39.1	293.2	5.1	10.1
Mean IS				8/9	27.5	−7.4	35.8	10.0						
Mean TC							36.8	9.9	25.4	13.6	41.1	288.1	$A_{95} = 8.4$	

Legend and explanation for this table: Bed is bedding measurements given as the strike and dip angle [down dip to the right (clockwise) of strike]. N indicates the ratio of samples used in the statistical analysis versus studied. Dec and Inc are the declination and inclination of the site-mean direction (in °); k is Fisher's (1953) concentration parameter; α_{95} is the radius of the 95 per cent confidence cone about the mean direction (in °). VGP is virtual geomagnetic pole; Lat is latitude; Lon is longitude; dp and dm are semi-minor and semi-major axes of oval of 95 per cent confidence. A_{95} is 95 per cent confidence limit for mean palaeomagnetic pole. IS, *in situ*; TC, tilt-corrected directions.
Note: [a]directions close to the local PDF; these have been excluded from the collection-mean.

in the magnetization directions at ∼500 °C. However, in all such cases, the direction of the intermediate temperature component lies on a great circle segment connecting the LTC and high temperature component (HTC) directions. Such a relationship is typical of composite components resulting from an incomplete separation of the LT and HT magnetizations because of the overlap in their blocking temperatures spectra. It is likely that the 'intermediate' direction has no physical meaning.

The HTC trending to the origin was isolated in all studied samples. Maximum unblocking temperatures of the component range between 560 and 620 °C, indicating that remanence is carried by (titano)magnetite and/or maghemite. The relatively large variations in the unblocking temperatures of the LT and HT components between the samples are likely because of the variations in the composition and grain size of carriers between different flows as well as differences in the cooling rate between the interior and margins of different flows (e.g. Beske-Diehl & Soroka 1984; Dunlop & Özdemir 1997).

The directions of the HTCs fall into two groups (Table 1; Fig. 6). The directions isolated at site Sz-15i as well as the HTC of two

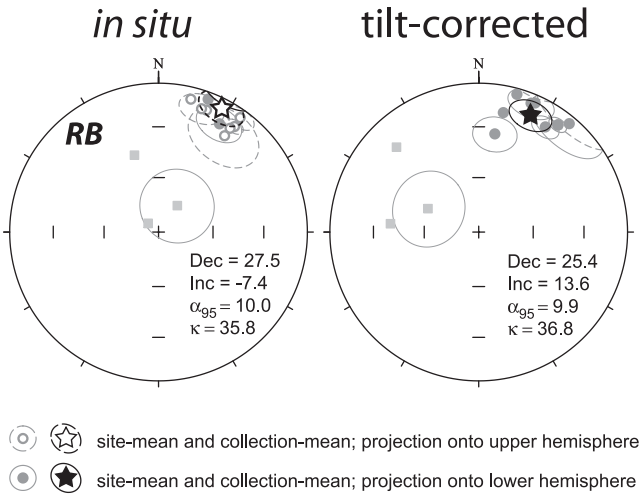


Figure 6. Equal-area stereoplots of site-mean magnetization directions of the RB collection (as also listed in Table 1).

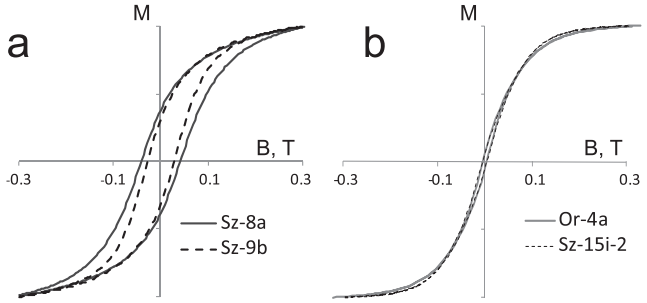


Figure 7. Representative examples of the hysteresis curves (corrected for paramagnetic contributions) for (a) samples with the characteristic direction, (b) rejected samples, remanence direction of which is close to the local geomagnetic field direction.

samples (Or-3 and Or-4) from another flow are somewhat scattered around the present day geomagnetic field direction (PDF) at the sampling locality (square symbols in Fig. 6, italics in Table 1). The shallow NE directions of the second group (shown by circles in Fig. 6) we designated as characteristic for this collection. Hysteresis measurements (Fig. 7) demonstrated significant differences in magnetic properties between the two groups. The samples with the PDF-like directions have a low ratio of remanent to saturation magnetization and very low coercivity values (Fig. 7b) that are indicative of multidomain magnetic carriers. Multidomain grains, being susceptible to acquisition of viscous magnetization (e.g. Dunlop & Özdemir 1997), can account for the recent (viscous) origin of remanence in this group of the samples.

In contrast, the samples with the characteristic magnetization (ChRM) have a high ratio of remanent to saturation magnetization and high coercivity values (Fig. 7a)—the features indicative of single-domain to pseudo-single-domain magnetic grain sizes. Such grains are known to be efficient in recording and retaining a primary magnetization (e.g. Dunlop & Özdemir 1997); accordingly, the presence of single-domain magnetization carriers in rocks argues in favour of primary origin of the remanence.

Unfortunately, there is no formal way to confirm the primary age of the ChRM. The little variation in the bedding attitudes of the basalt flows at the sampling locality does not allow for the fold

test; and no other conventional field tests were available. However, neither *in situ* nor tilt-corrected ChRM directions resemble any post-Cretaceous Eurasian reference (Torsvik *et al.* 2008) inclinations for the sampling locality. Thus, it is highly unlikely that the ChRM represent a post-Cretaceous remagnetization; the primary nature of the remanence is much more likely.

An ever-present problem in palaeomagnetic studies of lava sequences, especially those with a small number of sampled flows, is whether the primary remanence direction is representative of the geocentric axial dipole field direction. In some sequences that were emplaced during short bursts of volcanic activity, the lavas are known to record a transitional field direction (e.g. Coe & Prévot 1989), or the duration of the volcanic activity may be too short to capture the full range of the geomagnetic secular variations (e.g. Knight *et al.* 2004).

In our Rozhdestvenka collection, the presence of multiple chert beds, a slowly accumulating type of sediments, in between the basalt flows indicates that the sequence formed over a long time period, at least several tens to hundreds of thousands of years. The geomagnetic field changes, such as an excursion or a transition from one polarity to an opposite one, usually happen on a shorter timescale, typically in less than 10 000 yr (Merrill & McFadden 1999; Clement 2004; Roberts 2008). Given the similarity of the ChRM directions recorded in several flows of widely different age, it is highly unlikely that the remanence of all flows was acquired during a transitional field episode.

On the other hand, the remanence directions recorded in the sampled flows and their corresponding virtual geomagnetic poles (VGP) show a reasonable spread. We can use the angular dispersion of the VGPs about their mean, after correction for the dispersion because of random errors associated with sample orienting and measuring, as an estimate of palaeosecular variations (PSV; e.g. Cox 1970; Biggin *et al.* 2008). The PSV characteristics during the Cretaceous normal superchron have been discussed by Biggin *et al.* (2008) who have shown that at low latitudes, PSV might be expected to produce VGPs' dispersion values of $\sim 12^{o}$ for low quality (small number of points) data sets. The VGPs' dispersion of the Rozhdestvenka collection is 13.6 (for 8 flows), quite close to the expected. We also note that the stratigraphic succession of the VGPs in our collection does not show any consistent trend (Fig. 8). A consistent trend, produced as remanence directions track the drift of the geomagnetic pole, can be a symptom of a sequence of lavas

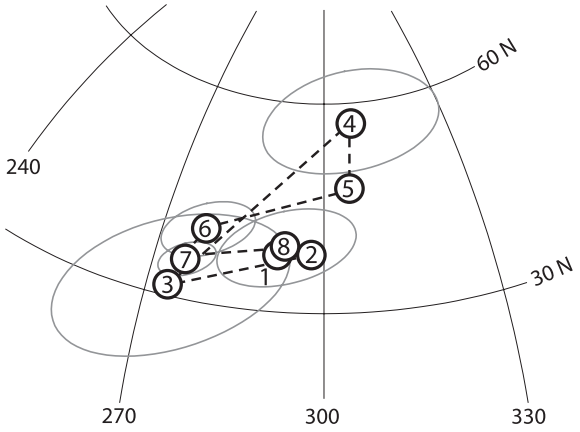


Figure 8. Virtual geomagnetic poles positions from Rozdestvenka basalt units shown by numbered circles (with 95 per cent confidence where applicable). Number within the circle indicates a sequential order of the basalt flow, listed from oldest to youngest.

under-representing PSV (e.g. Biggin *et al.* 2008); our collection lacks this signature.

Overall, we found that despite the absence of conventional field tests the balance of probabilities favours the primary origin for the ChRM of the Rozdestvenka basalts. Although there is no objective way to establish how well the PSV are averaged out in our collection, we have found no evidence to suggest that PSV might be underrepresented. In view of that, we consider that the mean direction of the ChRM of the Rozhdestvenka collection is representative of the axial geomagnetic field at the time of the formation of the basalts. As according to their biostratigraphically constrained Berriasian–Valanginian age (Zyabrev & Bragin 1987), the basalt flows formed during the Cretaceous normal superchron, the positive (tilt corrected) inclination of the ChRM indicates their formation in the Northern Hemisphere.

5.2 Naiba sequence

5.2.1 Rock magnetism

To evaluate the fidelity of palaeomagnetic recording (potentially suspect given the unstable demagnetization behaviour of the pilot specimens), we conducted a set of rock-magnetic experiments designed to identify the main magnetization carriers in the Naiba sequence sedimentary deposits. The characteristic magnetic properties are shown in Fig. 9. The thermomagnetic runs are irreversible for all samples (Fig. 9a), and the majority of the samples have a peak increase in magnetization at 400–600°C on the heating branch (Fig. 9a, N33–2). These features are indicative of mineralogical changes taking place in this temperature range. ZFC and FC warming curves have a distinctive Verwey transition at ∼100–120 K typical of magnetite (Fig. 9b). However, a characteristic shape of RT IRM on cooling—an initial rise in intensity followed by a gentle slope on approaching the Verwey transition (Fig. 9c)—indicates the presence of partially oxidized magnetite, or maghemite (Özdemir & Dunlop 2010).

Maghemite is a common mineral phase in the sedimentary rocks where it occurs either as detrital grains or forms *in situ* by partial oxidation of (titano)magnetite grains. In maghemite formed by partial oxidation, a primary remanence direction of the original magnetite grain is often preserved (Özdemir & Dunlop 1985). Overall, maghemite as a carrier of detrital remanence is considered to be a faithful recorder of the geomagnetic field. However, maghemite is metastable, and known to undergo mineralogical transformations at above 400 °C (Bagin *et al.* 1988; Dunlop & Özdemir 1997). The alteration of maghemite likely accounts for the acquisition of spurious magnetization by sedimentary rocks of the Naiba sequence during thermal demagnetization.

5.2.2 Palaeomagnetism

In sandstone samples of the Naiba sequence, two magnetization components were usually isolated either by the thermal demagnetization alone or by the composite treatment (Figs 4 and 10). A LTC is well defined up to 400 °C in all samples. With few exceptions, the direction of this component is close to that of the present day field at the studied locality. A HTC linearly decaying to the origin was isolated in the majority of the samples. Although in some samples 400 °C treatment completely removes the LTC (e.g. Figs 10b, g and f), in others the transition between the LTC and HTC is gradual,

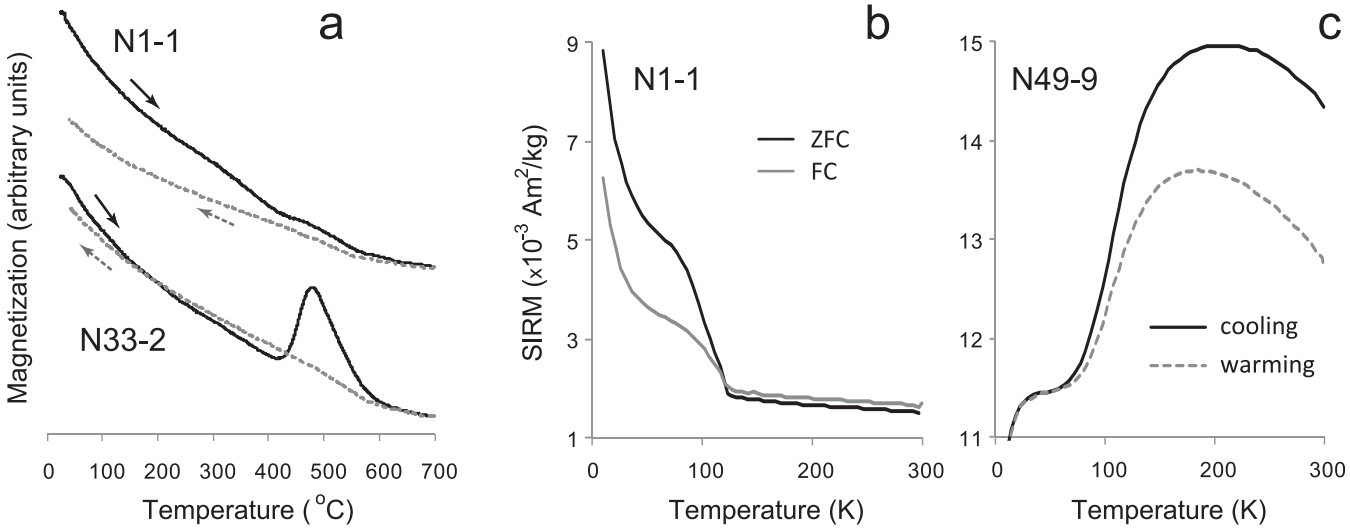


Figure 9. Rock magnetic characteristics of Naiba sequence sediments; (a) examples of thermomagnetic runs; (b) saturation remanence curves after a sample was cooled in a high field (FC) and a zero field cooled (ZFC); (c) room temperature remanence during cooling and subsequent warming.

with magnetization endpoints following curved demagnetization paths (e.g. Figs 10d and e).

Demagnetization behaviour of the conglomerate clasts is generally different from that of the studied sandstones—not surprising given the difference in lithology (Fig. 11). The pebbles reveal either a single dominant component (e.g. Figs 10a and d), or two magnetization components (e.g. Figs 10c and f). The directions of the HTCs pass the test for randomness at a 95 per cent confidence level (Watson 1956), with the length of the resultant HTC vector $\mathbf{R} = 4.23$ being smaller than the critical value of $\mathbf{R}_o = 9.09$ for 32 points. Notwithstanding the lithological differences, the positive conglomerate test suggests that studied section did not suffer a blanket remagnetization, and the primary magnetization is likely to be preserved in the sandstones as well.

With the exception of site N20 where we failed to isolate a ChRM, the directions of the HT component usually group reasonably well ($\alpha_{95} < 20^o$) on the site level (Table 2). On the collection level, we found a rather large spread in site-mean declinations, and for the statistical treatment, the collection was subdivided into four groups according to the biostratigraphic age of rocks.

The ChRM directions of the Ai Fm sites (Ai collection, Early Albian age) group better after tilt correction (Fig. 12). The bootstrap fold test of Tauxe & Watson (1994) indicates that the best grouping is achieved at 97 per cent unfolding with a 95 per cent confidence interval of 64–130 per cent. The positive fold test indicates a pre-folding age of the magnetization.

The small range of variation in the bedding attitudes of the sampled Late Albian-age sites (Lower Naiba Fm; LN collection) results in large confidence limits of 23.3–150.0 per cent of the Tauxe & Watson (1994) fold test. Although the test can be considered insignificant because of the wide confidence range, the best grouping of characteristic directions is nevertheless is achieved upon tilt correction, at 103.3 per cent unfolding. The tilt-corrected collection-mean direction is also close to the mean ChRM direction of the Ai Fm sites for which the pre-folding age of magnetization is supported by the more robust fold test.

Similarly, the small bedding variations of the Cenomanian-age sites (Upper Naiba–Lower Bykov Fm; UN–LB collection) result in a weak fold test with large confidence limits of 57.0143.8 per cent, and the best grouping at 97.3 per cent unfolding. However, the random ChRMs measured from individual cobbles in an intraformational conglomerate (Fig. 11) support a primary origin for the magnetic remanence in this part of the sedimentary sequence.

Site-mean directions of the twenty four sampled sites of the Krasnoyarka Fm (Campanian–Maastrichtian, Kr collection) fell into two antipodal groups that pass the reversal test of McFadden & McElhinny (1990) with the observed $\gamma = 6.34$ being smaller than the critical $\gamma = 7.53$. The directions also pass the Tauxe & Watson (1994) fold test with the best grouping achieved at 98.0 per cent unfolding (Fig. 12).

Overall, despite the difference in confidence limits on the fold tests, all collections of the Naiba sequence have better grouping of characteristic directions in tilt-corrected coordinates. The evidence for the pre-folding age of magnetization, the positive conglomerate test (Naiba Fm), the positive reversals test (Krasnoyarka Fm) suggest that the ChRM of the Naiba sequence is of primary origin.

5.2.3 Anisotropy of remanence

As a check for sediment compaction and possible inclination shallowing, we measured AARM for 16 samples representative of eight sites of the Ai Formation; the results are summarized in Fig. 13. Samples from two sites, N-2 and N-3, have a well-developed lineation in northwest–southeast direction (Figs 13a and b). This direction is similar to the orientation of the flute casts measured in this part of the section (Zyabrev 1992); thus, the ARM fabric likely resulted from primary current directions.

The rest of the samples have low anisotropy (maximum/minimum axes), usually below 5 per cent (Fig. 13a), with a wide scatter of the minimum ellipsoid axes (Fig. 13c). Overall, our samples lack the hallmark signature of compaction—a well-developed foliation with the minimum axes (ARM_{min}) perpendicular to bedding (e.g. Jackson *et al.* 1991). We also did not find any correlation between the observed inclination of characteristic remanence (tan I) and fabric intensity (Fig. 13d), which is expected theoretically for a suite of samples with inclination shallowing (Hodych & Bijaksana 1993). Therefore, we conclude that no significant inclination error is present in Ai Fm sandstone samples and no correction for anomalously shallow inclinations is necessary.

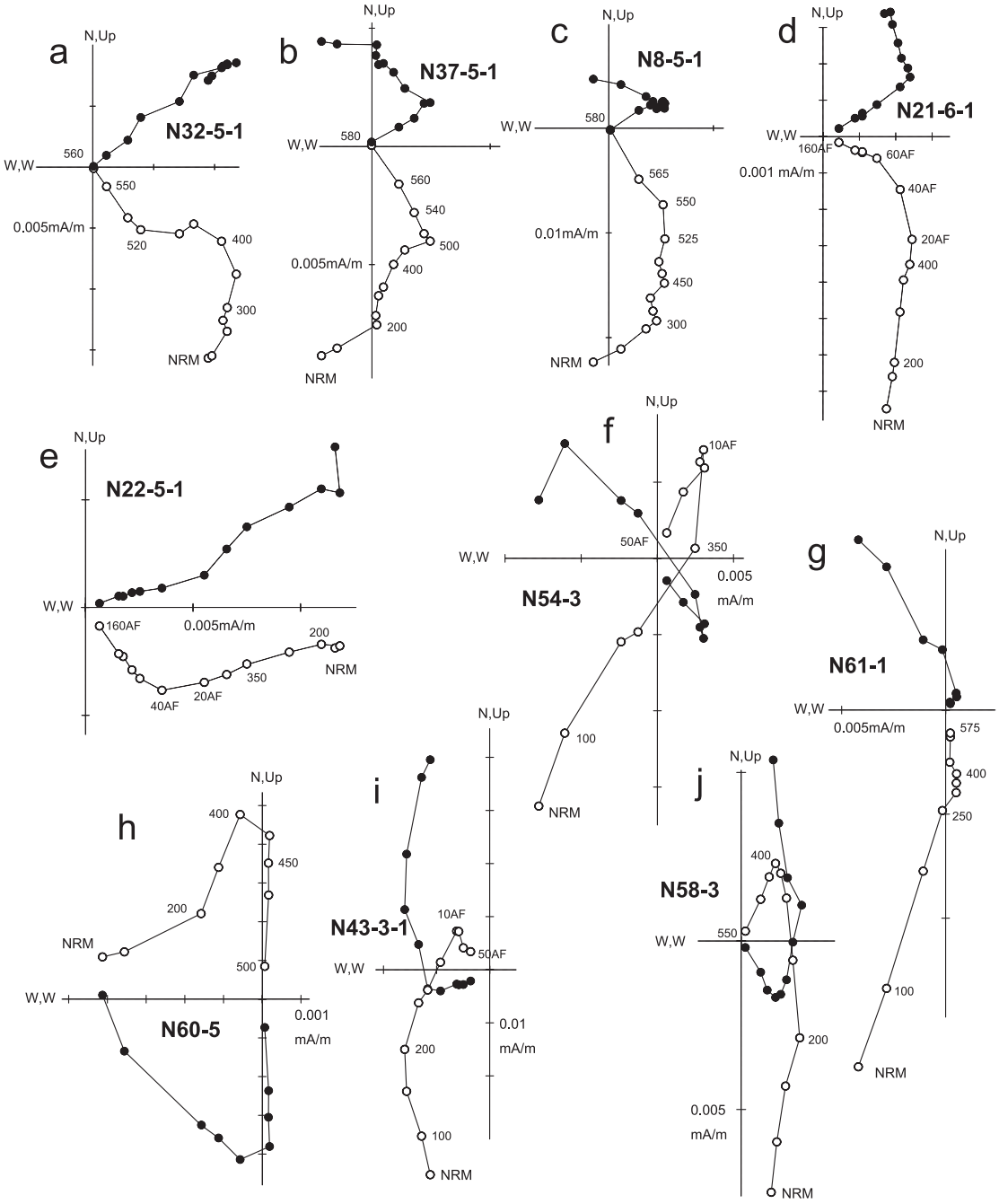


Figure 10. Orthogonal demagnetization diagrams in tilt-corrected coordinates for representative samples of the Naiba sequence; (a, b) Ai Fm; (c) Naiba Fm; (d, e) Bykov Fm; (f, g, h, i, j) Krasnoyarka Fm. Plotting conventions as in Fig. 4.

6 DISCUSSION

6.1 Latitudinal displacement

The WSB along its entire north–south extent is considered to be a single tectonic element with continuously traceable lithostratigraphic units (Zyabrev 1992 and references therein). An east–west trending fault to the south of our Rozdestvenka locality (locality 1 in Fig. 1a) is a Cenozoic transcurrent fault with a limited dextral displacement that truncates and slightly offset the early structural plan (Zyabrev 1992). There is no evidence for north–south shortening (i.e. for the relative latitudinal displacement between our two sampling localities) associated with this fault, or anywhere else within the WS basin. Accordingly, the palaeolatitudes estimates obtained from the two localities can be extended to the entire basin. In the following discussion, Table 3 and Fig. 14(a), the palaeolatitudes are calculated for a common point at 47.32°N, 142.56°E (the Naiba area).

The progression of the Naiba area latitudes (Table 3; Fig. 14a) show a gradual shift of the WSB from subequatorial latitudes during the Early Cretaceous to about 40°N by the Late Cretaceous. The Early Cretaceous latitudes (for basalt of the basement) are ∼28 degrees lower than those predicted by extrapolation from the mean Stable Eurasia poles (Torsvik *et al.* 2008; Fig. 14a; Table 3). As

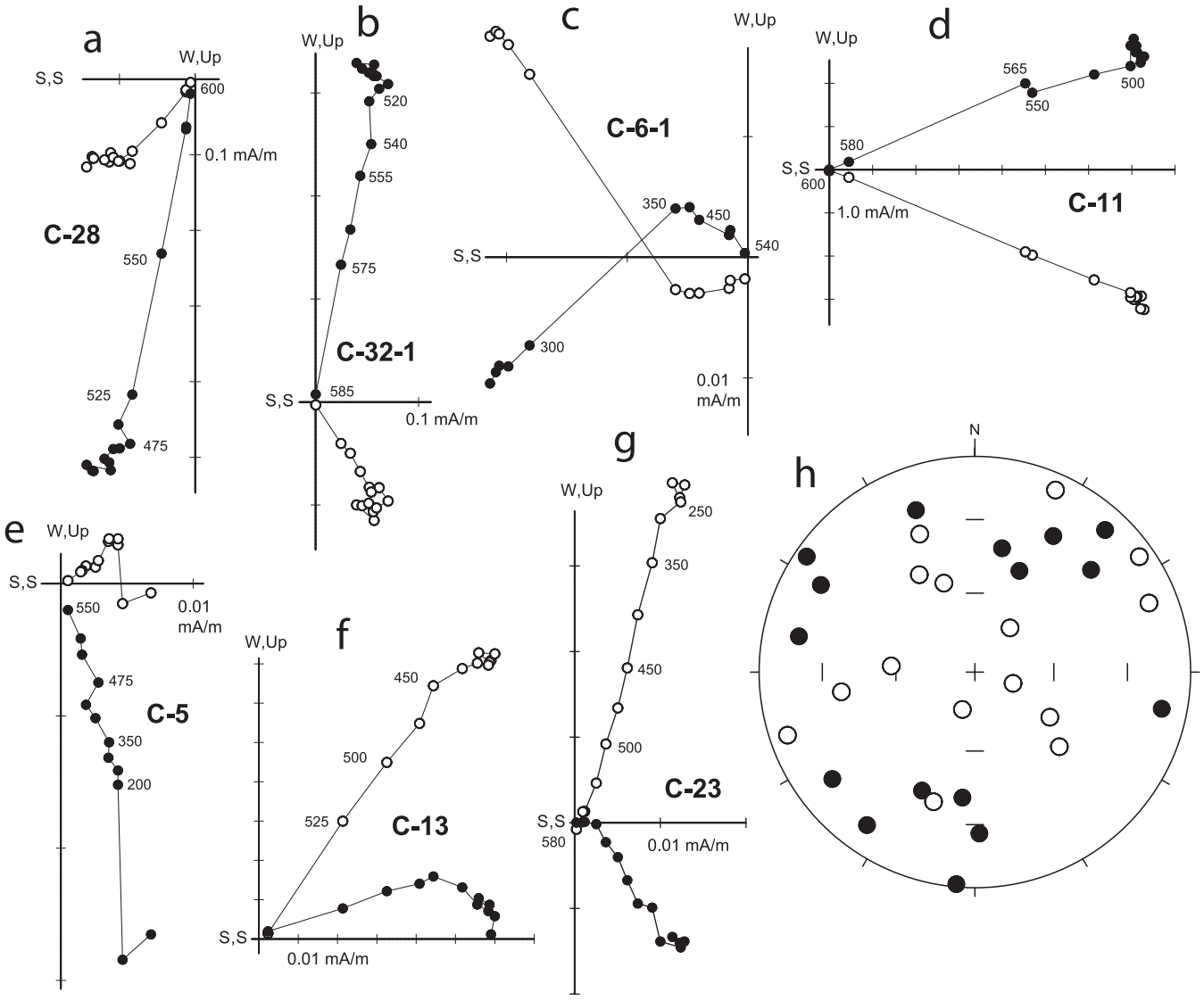


Figure 11. Examples of demagnetization diagrams (*in situ*) of conglomerate clasts (a–g); plotting conventions as in Fig. 4. Stereographic projection of the characteristic directions of conglomerate clasts (h); solid symbols denote projection onto the lower hemisphere, open ones are projections onto the upper hemisphere.

the poles obtained from the volcanic rocks are not subject to the inclination shallowing, the RB palaeopole provides a robust estimate for the Early Cretaceous location of the basin's basement.

The statistically significant (i.e. the confidence levels of the predicted and observed mean estimates do not overlap) discrepancy of $\sim 11 \pm 6^{o}$ is found for the Early Albian (Ai Fm) palaeolatitudes. These low palaeolatitudes inferred for the volcanic rocks and sediments of the WSB at the time of their formation cannot be explained by the compaction-induced inclination shallowing as the measurements of the remanence anisotropy do not show evidence for a well-developed compaction fabric in the studied rocks. Moreover, the magnetic inclinations of other formations that are similar to the Ai Fm deposits in composition and grainsize (i.e. factors controlling the compaction and, consequently, the inclination shallowing, e.g. Dickinson & Butler 1998) do not show a shallow bias.

The Late Albian palaeolatitudes of the basin (based on the Lower Naiba Fm palaeopole) are slightly lower than that predicted from the stable Asian continent. Although the confidence limits of the observed and predicted directions partially overlap at the limits of assigned error, they do not include respective mean values. Such a distribution suggests that the difference in mean latitudes is statistically meaningful.

Since the Cenomanian, the observed palaeolatiudes are statistically indistinguishable from the predicted ones (Fig. 14a). Accordingly, the main phase of the northward migration of the WSB can be constrained to the Early Cretaceous, and it was likely finished by the Cenomanian. This timing of the termination is consistent with the previous study of the Yezo Group deposits by Tamaki & Itoh (2008) that did not detect any significant migration of the basin since the Cenomanian.

The average latitudinal migration velocity thus can be estimated at ~ 7 cm yr^{-1}. Although comparable with the modern rate of the Pacific Plate motion (DeMets *et al.* 2010), this value is significantly lower than the estimated velocity of the Izanagi Plate (~ 20.5 cm yr^{-1}; Maruyama & Seno 1986) that was bordering the eastern Asian margin during the Early Cretaceous.

Table 2. Characteristic remanence directions of the Naiba sequence.

Site	Location		Bed	N	*In situ*			Tilt corrected			VGP			
	Lat (N)	Long (E)			Dec	Inc	k	α_{95}	Dec	Inc	Lat (N)	Lon (E)	dp	dm
Ai Fm (Ai)														
N1	47.31	142.59	178/40	4/6	73.1	10.1	24.4	18.9	63.1	47.8	38.6	229.9	16.1	24.6
N2	47.31	142.59	188/39	6/7	75.9	8.5	40.0	10.7	67.0	43.8	33.8	230.2	8.4	13.4
N3	47.31	142.59	175/34	6/6	63.5	10.6	123.0	6.1	56.1	41.6	40.1	240.1	4.6	7.5
N32	47.31	142.59	125/56	4/5	66.8	−2.4	24.5	18.9	80.9	42.9	24.0	221.2	14.5	23.4
N33	47.31	142.59	125/56	4/5	49.9	−6.4	32.4	16.4	57.1	47.2	42.3	234.9	13.8	21.2
N34	47.31	142.59	150/40	5/5	44.0	4.8	58.9	10.1	37.9	42.9	52.6	255.9	7.7	12.5
N35	47.31	142.59	150/39	6/6	58.6	23.3	24.6	13.8	57.2	62.3	50.6	216.2	16.7	21.5
N36	47.31	142.59	171/57	5/5	56.4	−7.7	68.9	9.3	46.6	43.1	47.2	247.1	7.2	11.5
N37	47.31	142.59	171/70	5/5	52.6	−30.7	27.5	14.8	52	32.4	38.3	249.4	9.4	16.7
N38	47.31	142.59	175/70	5/6	66.6	−12.4	128.2	6.8	54.2	52.9	47.4	231.3	6.5	9.4
Mean IS				10/10	60.8	−0.2	20.0	11.1						
Mean TC				10/10			50.8	6.8	57.1	46.2	42.1	235.1	$A_{95} = 7.5$	
Lower Naiba Fm (LN)														
N4	47.31	142.59	168/32	7/7	70.6	14.1	29.9	11.2	67.7	45.7	34.3	228.4	9.1	14.3
N5	47.31	142.59	165/38	6/6	64.0	14.0	34.6	11.5	57.9	5.1	43.8	230.7	10.5	15.5
N6	47.30	142.59	170/60	6/6	65.0	0.8	30.0	12.4	51.2	57.5	52.0	227.3	13.3	18.2
N7	47.30	142.59	181/58	6 / 6	56.1	−5.2	33.2	11.8	42.8	40.1	48.1	253.1	8.6	14.2
N8	47.30	142.59	180/56	6/6	70.2	4.4	47.9	9.8	53.8	55.1	48.9	228.9	9.9	13.9
N9	47.30	142.59	178/59	6/6	59.7	6.4	33.8	11.7	35.0	53.9	60.9	246.0	11.5	16.4
N10	47.30	142.59	177/60	5/6	67.5	−3.5	27.5	14.9	54.5	51.7	46.5	232.4	13.8	20.3
N11	47.30	142.59	174/55	5 / 6	42.1	2.5	40.0	12.2	24.4	39.3	58.2	276.1	8.7	14.6
N12	47.30	142.59	178/57	5/5	80.2	10.9	180.3	5.7	68.1	66.8	46.3	203.5	7.8	9.4
N30	47.30	142.59	177/72	5/5	76.0	−9.6	86.0	8.3	64.7	60.3	44.6	215.4	9.6	12.6
N31	47.30	142.59	180/63	5/6	77.5	−12.1	28.1	14.7	71.1	49.1	33.9	223.4	12.9	19.4
Mean IS				11/11	66.3	2.1	33.2	8.0						
Mean TC							42.3	7.1	52.6	52.8	48.4	231.3	$A_{95} = 8.4$	
Upper Naiba Fm–Lower Bykov Fm (UN–LB)														
N13	47.32	142.57	184/70	4/6	71.8	10.6	43.9	14.0	24.4	66.7	73.7	216.7	19.1	23.1
N14	47.32	142.57	184/70	5/6	82.0	3.8	101.0	7.7	56.6	70.0	54.5	200.3	11.4	13.2
N15	47.32	142.57	185/68	3/6	62.7	−7.8	224.7	8.2	44.7	46.5	50.3	245.8	6.8	10.5
N16	47.32	142.57	192/78	6/6	67.0	−3.5	40.7	10.6	33.9	51.9	60.4	250.3	9.9	14.5
N17	47.32	142.57	187/87	6/6	75.1	−12.2	43.3	10.3	42.3	63.5	61.0	221.1	12.9	16.3
N18	47.32	142.57	190/81	7/7	70.9	2.2	45.6	9.0	21.4	60.3	73.6	246.5	10.4	13.7
N20	47.30	142.57	186/63	0/6	–	–	–	–	–	–				
N21	47.30	142.57	186/63	5/6	58.8	6.4	56.2	10.3	29.2	50.8	62.7	257.4	9.4	13.9
N22	47.30	142.57	184/62	6/6	57.5	5.5	20.4	15.2	30.2	48.7	60.8	258.9	13.2	20.0
N23	47.30	142.57	187/63	6/6	66.1	−2.2	43.9	10.2	46.5	48.3	50.1	242.4	8.8	13.4
N24	47.30	142.57	177/62	5/6	66.8	1.2	85.3	8.3	47.8	57.0	54.0	230.4	8.8	12.1
N25	47.30	142.57	183/60	4/6	50.7	15.7	33.2	16.2	13.7	48.8	69.5	286.6	14.1	21.4
N26	47.31	142.57	177/65	4/6	61.5	−5.9	24.6	18.9	44.8	50.4	52.4	241.6	17.0	25.4
N27	47.31	142.57	177/65	5/5	61.0	−3.3	55.9	10.3	41.6	52.1	55.5	242.3	9.7	14.1
N28	47.31	142.57	175/64	4/5	50.7	8.9	33.9	16.0	16.4	53.2	71.7	274.3	15.4	22.2
N29	47.31	142.57	175/65	5/5	58.4	1.9	104.5	7.5	33.9	54.9	62.2	245.5	7.5	10.6
Mean IS				15/16	64.1	1.4	51.0	5.4						
Mean TC				15/16			65.9	4.7	34.8	55.4	62.2	242.4	$A_{95} = 6.0$	
Krasnoyarka Fm (Kr)														
N39	47.32	142.50	183/39	6/6	56.4	41.0	43.8	10.3	10.3	63.0	82.3	250.5	12.7	16.2
N40	47.32	142.50	183/39	6/6	59.7	37.5	36.3	11.3	19.9	62.9	75.9	237.4	13.9	17.7
N41	47.32	142.50	183/39	6/6	50.3	30.2	41.7	10.5	19.8	52.2	69.1	269.4	9.9	14.4
N42	47.32	142.50	183/39	6/6	57.0	44.0	52.0	9.4	5.0	65.0	86.6	236.1	12.2	15.2
N43	47.33	142.50	188/43	5/6	232.0	−39.9	37.9	9.9	183	−56.4	79.4	309.4	10.3	14.3
N44	47.33	142.50	186/38	7/7	40.9	24.7	32.8	11.9	17.5	40.5	62.0	286.4	8.7	14.4
N45	47.33	142.50	185/38	6/6	46.0	38.0	24.1	13.9	7.7	53.5	75.5	296.2	13.5	19.4
N46	47.33	142.50	188/36	6/6	59.8	37.8	38.3	11.0	24.2	59.4	71.2	245.9	12.3	16.5
N47	47.33	142.51	290/42	6/6	261.0	78.3	27.0	13.1	3.1	52.5	75.6	312.0	12.4	18.0
N48	47.33	142.51	290/42	6/6	182.7	74.1	81.0	7.5	30.3	62.8	68.8	229.7	9.2	11.8
N49	47.33	142.51	290/42	6/6	191.0	68.4	89.4	8.1	29.3	69.2	70.6	205.4	11.8	13.8
N50	47.33	142.51	290/42	5/6	208.7	70.2	91.9	9.6	12.1	67.4	81.5	208.1	13.3	16.0
N51	47.33	142.51	177/34	4/6	41.3	46.5	35.9	12.9	356.3	60.5	83.6	348.1	14.9	19.6
N52	47.33	142.51	165/25	5/5	29.7	51.3	162.0	6.0	353.7	63.3	85.0	25.0	7.5	9.5

Table 2. (*Continued.*)

Site	Location		Bed	N	*In situ*			Tilt corrected			VGP			
	Lat (N)	Long (E)			Dec	Inc	k	α_{95}	Dec	Inc	Lat (N)	Lon (E)	dp	dm
N53	47.34	142.52	162/28	5/5	215.7	−41.6	58.1	10.1	188.3	−60.4	81.6	274.8	11.7	15.3
N54	47.34	142.52	162/28	6/6	214.3	−45.1	19.1	15.7	182.3	−62.6	86.3	296.2	19.2	24.5
N55	47.34	142.52	162/28	6/6	209.2	−45.4	24.7	13.7	176.2	−60.5	83.5	348.7	15.9	20.8
N56	47.34	142.52	162/28	6/6	220.1	−41.0	36.6	11.2	194.9	−61.7	78.6	250.1	13.4	17.3
N57	47.34	142.52	162/30	5/6	217.7	−31.4	71.6	9.1	198.1	−53.5	71.0	270.3	8.8	12.7
N58	47.34	142.52	157/22	5/6	186.0	−51.5	50.3	10.9	155.6	−57.0	69.7	32.8	11.5	15.9
N59	47.34	142.52	157/22	6/6	206.3	−30.0	34.2	11.6	193.7	−45.2	66.7	290.1	9.3	14.7
N60	47.34	142.52	157/22	5/5	204.7	−43.1	42.8	11.8	183.5	−56.7	79.6	306.9	12.4	17.1
N61	47.34	142.52	157/22	6/6	44.4	50.2	61.4	8.6	23.6	69.0	74.0	205.0	12.4	14.6
N62	47.34	142.52	157/22	5/6	49.1	30.0	62.4	9.8	42.2	50.6	54.2	243.7	8.9	13.2
Mean IS				24/24	42.9	50.2	9.4	10.2						
Mean TC				24/24			63.5	3.7	11.3	59.3	79.6	267.5	$A_{95} = 4.8$	

Note: Notations are as in Table 1.

6.2 Rotations

Palaeomagnetic investigations of vertical axis rotations have been an important component of tectonic studies in the region for many years. The pattern of rotations have been used in reconstructing the Cenozoic kinematic evolution of the NE Asian margin, including the backarc opening of the Japan Sea basin and a regional reorganization of interacting continental and oceanic blocks (e.g. Jolivet *et al.* 1994; Worrall *et al.* 1996; Takeuchi *et al.* 1999), as well as in delineating presently poorly constrained boundaries between the Pacific, North American, Eurasian and Okhotsk Sea plates (e.g. Seno *et al.* 1996; Fournier *et al.* 1994; Weaver *et al.* 2003).

Significant vertical axis rotations, both clockwise and counterclockwise, have been previously reported from Sakhalin and the adjacent areas (e.g. Takeuchi *et al.* 1999; Weaver *et al.* 2003; Otofuji *et al.* 2003, 2006). These rotations have been attributed to Cenozoic activity on strike-slip faults associated with the opening of the Japan Sea and Kuril Basin, to local tectonic deformations associated with accretion of exotic terranes and to the present activity on the major strike-slip faults (Weaver *et al.* 2003; Otofuji *et al.* 2006; Tamaki & Itoh 2008). Declination deviations observed in the Cretaceous rocks in the area are usually discussed in the context of the Cenozoic deformations as well (e.g. Otofuji *et al.* 2006; Tamaki & Itoh 2008). Our result extends the history of significant vertical axis rotations in Sakhalin into mid-Cretaceous times.

In the continuous Naiba section, the declination deviations calculated relative to the Eurasian reference of Torsvik *et al.* (2008) progressively decrease with the age of rocks; from ∼66° in the Early Albian (Ai Fm) to ∼18° in the Campanian–Maastrichtian (Krasnoyarka Fm; Table 3; Fig. 14b). This decrease signifies ∼48° of total clockwise rotation during the Late Cretaceous (∼25 Ma interval). We note here that our estimate for the sense and magnitude of the rotation during the Cretaceous is obtained as a relative value, as the difference in the declination deviations values between the top and the bottom of a single continuous section. Consequently, the complex history of the Cenozoic block rotations in our sampling area discussed previously by Takeuchi *et al.* (1999) and Weaver *et al.* (2003) does not affect our findings.

To the best of our knowledge, this is the first study in the north–west Pacific region that explicitly demonstrated a large scale tectonic rotation occurring during the Late Cretaceous. From a single datapoint, it is not possible to deduce whether this rotation is related to the internal deformation of the basin, such as strike-slip associated block rotations, similar to the Cenozoic deformation style in the region (e.g. Takeuchi *et al.* 1999; Weaver *et al.* 2003; Otofuji *et al.* 2003, 2006); or is related to the rotation of the entire basin. Nevertheless, the documented rotation testifies that the deformation pattern of the WSB during the Cretaceous included block rotations with the magnitudes comparable to those of the Cenozoic (e.g. Weaver *et al.* 2003). The common approach to regard declination deviations observed in Cretaceous rocks of the region only in the context of the Cenozoic deformations (e.g. Otofuji *et al.* 2006; Tamaki & Itoh 2008) should be reevaluated.

6.3 Palaeogeography

In its present day location, the WSB lies approximately opposite to its coeval Cretaceous accretionary complexes of the mainland Asia (Fig. 1). Structure and distribution of the subduction related complexes suggests that in the Late Aptian–Early Albian, this part of margin must have had an active subduction zone, and thus had been opened to the ocean (e.g. Zyabrev *et al.* 2005). Based on this consideration, all tectonic models for the evolution of the East Asia postulate the post-mid-Cretaceous displacement of the WSB, albeit with no strict constraints on the timing and magnitude of displacement. Our palaeomagnetic study provides the first quantitative estimate of the palaeoposition of the WSB during the Cretaceous. Our results indicate that in the Early Albian (Ai Fm palaeople) the WSB was located by ∼1200 ± 500 km southwards from its final position relative to the Asian margin. A possible palaeogeographic reconstruction for this period, consistent with the available tectonic data, is shown in Fig. 15.

A detailed discussion of multiple tectonic constraints on timing and magnitude of displacements on major faults in the region is beyond the scope of this paper and the reader is encouraged to consult Natal'in (1993); Zyabrev *et al.* (2005), and references therein for thorough descriptions. In the following, we just outline a general procedure used for the reconstruction of the Early Albian the palaeogeography shown in Fig. 15. To restore the positions of the continental margin terranes, first, the displacements along the left-lateral Central Sikhote-Alin strike-slip fault and its parallel faults, as well as displacements associated with the opening of the Sea of Japan, were undone. Following this step, the displacements along the left-lateral Fushun-Mishan Fault were undone as well. The position of the WSB was inferred from the following considerations. Since the Cenomanian, the WSB, as a part of the

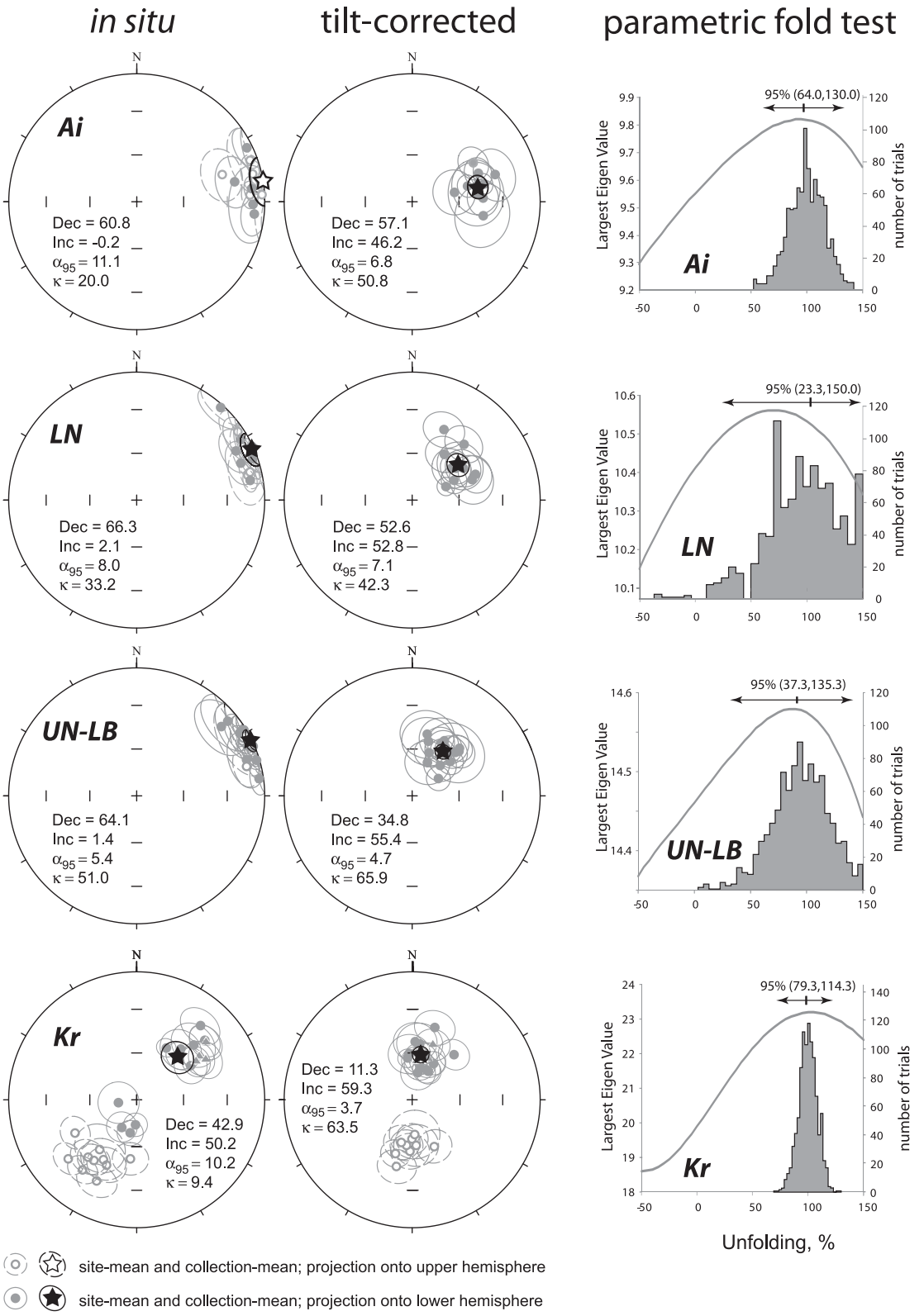


Figure 12. Equal-area stereoplots of site-mean magnetization directions of the Naiba sequence collections (as also listed in Table 2) and results of the parametric fold tests (Tauxe & Watson 1994) for the corresponding collections.

Cretaceous accretionary system paired with the East Sikhote-Alin volcanic belt ('esa' in Fig. 1a), was situated along the East Asian continental margin (Zyabrev 1992). To restore the WSB location during the Early Albian, it was back shifted to the south along the Asian continental margin, in accordance with the inferred northward motion of Izanagi oceanic plate (Maruyama & Seno 1986) by ∼11° of latitude (∼1200 km), in accordance with our Ai Fm result (Table 3).

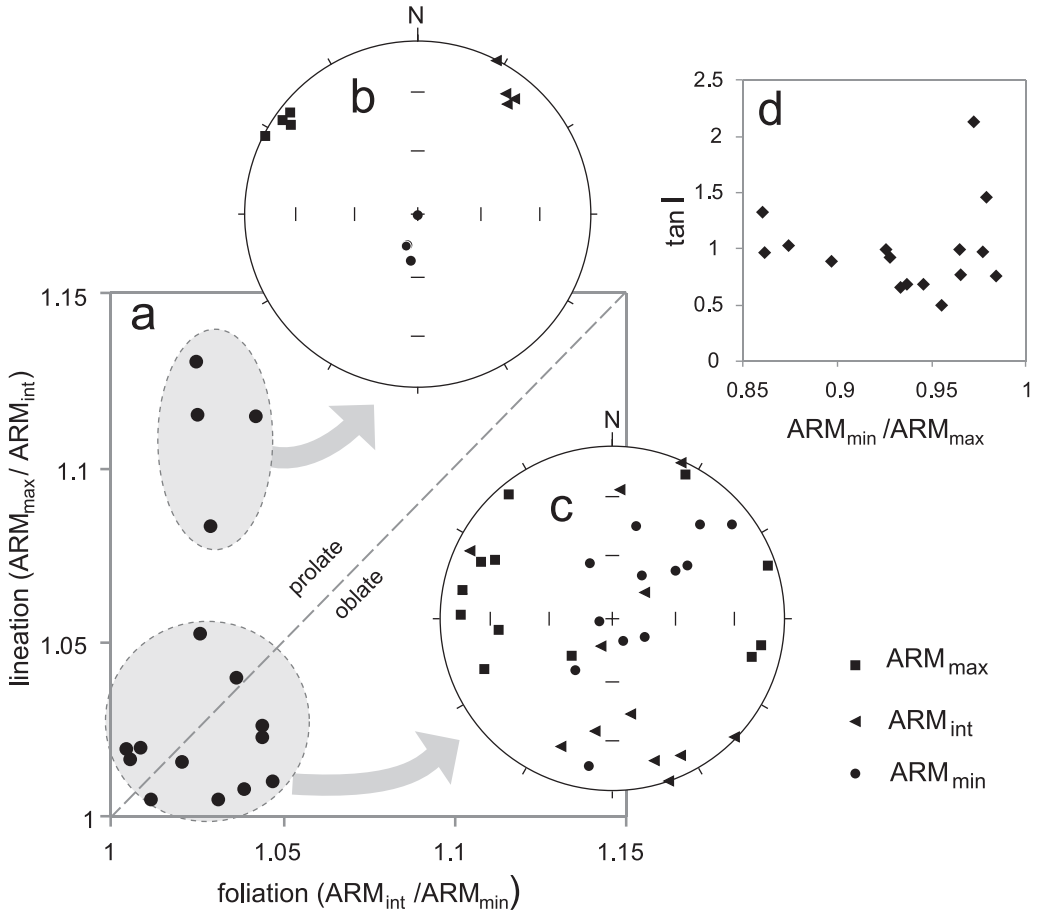


Figure 13. Anisotropy of remanence. (a) Flinn-type diagram of magnetic fabric; (b, c) equal area, lower hemisphere projection of principal ARM ellipsoid axes, in tilt-corrected coordinates; (d) inclination of characteristic remanence versus degree of flattening.

Table 3. Mean palaeopoles and their departure from the global APWP in European coordinates.

Collection	Age	P-lat	P-lon	A_{95}	λ at Naiba	Δλ	L	ΔL	R	ΔR
RB	Berriasian–Valanginian	41.1	288.1	8.4	3.6	$^{+7.7}/_{-7.8}$	27.7	$^{+9.5}/_{-9.6}$	36.9	9.8
Ai	Early Albian	42.1	235.1	7.5	28.1	$^{+4.6}/_{-5.0}$	10.8	$^{+6.0}/_{-6.3}$	66.3	7.2
LN	Late Albian	48.4	231.3	8.4	34.0	$^{+4.6}/_{-5.1}$	4.9	$^{+6.0}/_{-6.4}$	58.1	8.0
UN–LB	Cenomanian	62.2	242.4	6.0	36.6	$^{+3.2}/_{-3.5}$	2.0	$^{+4.0}/_{-4.2}$	39.8	6.3
Kr	Campanian–Maastrichtian	79.6	267.5	4.8	40.8	$^{+3.1}/_{-3.4}$	−2.2	$^{+3.9}/_{-4.2}$	18.4	4.8

Legend and explanation for this table: P-Lat, P-Lon are latitude and east longitude of the palaeopole; A_{95} is radius of 95 per cent confidence circle; λ is palaeolatitude, calculated for Naiba area at 47.32N, 142.56E; Δλ is confidence of palaeolatitude estimate; L is the difference between the observed and predicted (from global APWP in Eurasian coordinates of Torsvik *et al.* 2008) latitudes for Naiba area; ΔL is confidence level for latitudinal difference estimate; R is declination deviation from the predicted value with the corresponding 95 per cent confidence level ΔR. The uncertainty estimates for ΔL and ΔR were calculated using conventional error propagation.

7 CONCLUSIONS

Our palaeomagnetic study of the WSB rocks yielded several palaeopoles. A ChRM was isolated in 22 samples representative of eight individual flows of the RB (oceanic basement of the basin). No field tests were available to constrain the age of this magnetization. However, because it does not resemble any post-Cretaceous reference, we regard the ChRM direction as primary, dated to the Berriasian–Valanginian age of basalt emplacement. A palaeopole calculated from this direction is located at 41.1°N, 288.1°E ($A_{95} = 8.4$).

ChRMs have been isolated at 60 (out of 61) sites sampled from the sedimentary basin fill in the Naiba River area. A combination of the positive fold, conglomerate and reversals tests indicates that the ChRM is of primary origin. For statistical analysis, the sampled collection was subdivided into four age groups that yielded the following palaeopoles:

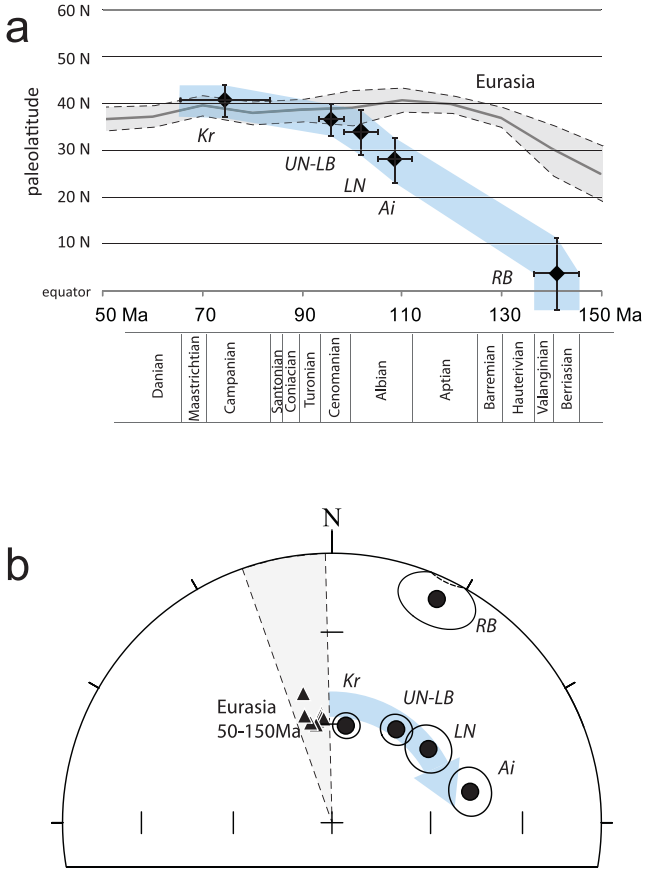


Figure 14. Summary of the results. (a) Plots of palaeolatitudes versus age for studied collections. All observed and reference palaeolatitudes are calculated for a common point at 47.32°N 142.56°E (The Naiba River area); the reference palaeolatitude curve with its associated confidence envelope is obtained by extrapolation from the Global APWP in European coordinates (Torsvik *et al.* 2008). Collections are indicated using the same abbreviations as in Figs 7 and 10. (b) Reference directions extrapolated from the Global APWP (black triangles, plotted in 10 Myr intervals, confidence limits are omitted for clarity) and directions of magnetization of the studied rocks. Filled circles are the mean magnetization directions, concentric ovals are cones of 95 per cent confidence. All directions are in the lower hemisphere, corresponding to normal-polarity field directions.

Early Albian (Ai Fm)	42.1°N, 235.1°E ($A_{95} = 7.5$)
Late Albian (Lower Naiba Fm)	48.4°N, 231.3°E ($A_{95} = 8.4$)
Cenomanian (Upper Naiba–Lower Bykov Fm)	62.2°N, 242.4°E ($A_{95} = 6.0$)
Campanian–Maastrichtian (Krasnoyarka Fm)	79.6°N, 267.5°E ($A_{95} = 4.8$)

The palaeopoles' progression indicates a gradual shift of the WSB from subequatorial latitudes during the Early Cretaceous to about 40°N by the Late Cretaceous. A comparison of the observed directions with the expected for Stable Eurasia indicates that the main phase of the margin-parallel migration of the WSB took place during the Early Cretaceous—the time of the oceanic basement development. In the Early Albian, soon after the commencement of siliciclastic sedimentation, the WSB still was ∼1200 ± 500 km southwards from its final destination. The northward migration of the terrane was likely finished by the Cenomanian. The continuous Naiba sedimentary sequence also records ∼50 degrees of progressive clockwise rotation during the Early Albian to Coniacian time.

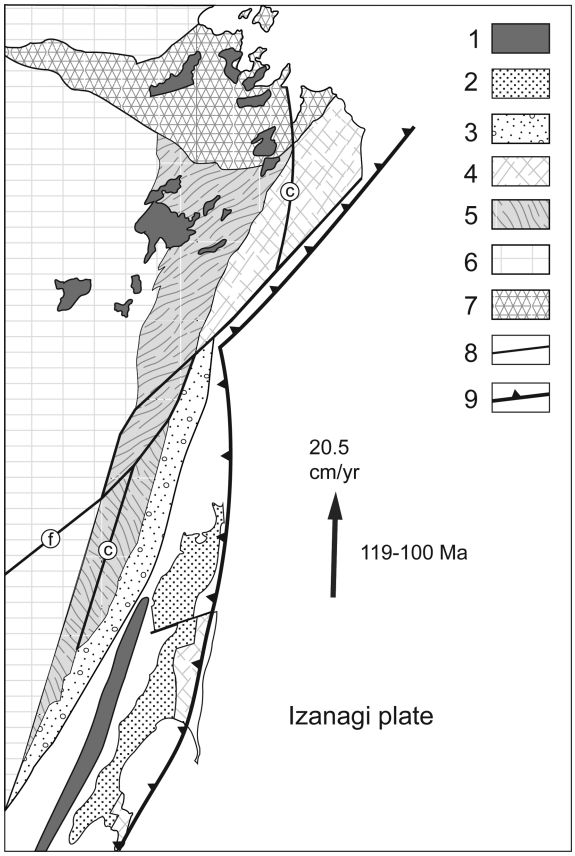


Figure 15. Palaeotectonic reconstruction of the east Asian margin in the Early Albian. The directions and rates of the movement of the Izanagi Plate relative to Eurasia according to Maruyama & Seno (1986). (1) Early–Late Cretaceous subduction-related volcanic belts; (2) forearc trough of West Sakhalin; (3) Early Cretaceous turbidite trough; (4) Cretaceous acccretionary complexes; (5) Jurassic–Early Cretaceous accretionary complexes; (6) cratonic areas; (7) Mongol–Okhotsk suture; (8) large faults, including traces of future Central Sikhote-Alin (c) and Fushun–Mishan (f); (9) subduction zone. Modified after Zyabrev (2011).

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